1. What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses artificial neural networks with many layers to learn patterns from large amounts of data automatically.

It is inspired by how the human brain processes information.

Example uses:

Image recognition

Speech recognition

Self-driving cars

Chatbots

2. What is a Neural Network?

A Neural Network is a computational model made of connected nodes (neurons) that process data in layers.

Layers include:

Input Layer

Hidden Layers

Output Layer

Each neuron processes input and passes output to the next layer.

3. Difference Between Machine Learning and Deep Learning

Machine Learning

-Subset of AI

-Requires manual feature extraction

-Works well with small datasets

-Uses simple algorithms

-Faster training

Deep Learning

-Subset of Machine Learning

-Automatically extracts features

-Requires large datasets

-Uses neural networks with many layers

-Requires more computation

4. What is Forward Propagation?

Forward Propagation is the process where:

Input data passes through the neural network

Each layer processes the data

The final output (prediction) is generated.

5. What is Backward Propagation?

Backward Propagation (Backpropagation) is the process of:

Calculating the error

Sending the error back through the network
Updating weights to improve the model.

6. What is Gradient Descent?
Gradient Descent is an optimization algorithm used to minimize the loss function by updating model parameters (weights).
It moves in the direction of the steepest decrease of error.

7. What is an Activation Function?
An Activation Function decides whether a neuron should be activated or not by transforming the input.
It introduces non-linearity into the network.

8. Types of Gradient Descent (Brief)
1. Batch Gradient Descent
Uses entire dataset
Stable but slow
2. Stochastic Gradient Descent (SGD)
Uses one data point at a time
Faster but noisy updates
3. Mini-Batch Gradient Descent
Uses small batches of data
Most commonly used

9. Types of Activation Functions

Sigmoid

Tanh

ReLU (Rectified Linear Unit)

Leaky ReLU

Softmax

10. What is a Loss Function?

A Loss Function measures how far the predicted value is from the actual value.

Example:

Mean Squared Error (MSE)

Cross Entropy Loss

Lower loss = Better model.

11. Difference Between Gradient Descent and SGD

Gradient Descent	Stochastic Gradient Descent
Uses full dataset	Uses one sample
Slow but stable	Fast but noisy
High memory usage	Low memory usage

12. Life Cycle of Deep Learning

Problem Definition

Data Collection
Data Preprocessing
Model Selection
Model Training
Model Evaluation
Model Deployment
Monitoring and Improvement

13. Applications of Deep Learning
Image recognition
Speech recognition
Natural Language Processing
Self-driving cars
Medical diagnosis
Fraud detection
Chatbots
Recommendation systems

14. What is Perceptron?
A Perceptron is the simplest type of neural network consisting of a single neuron used for binary classification.

15. What is Multi-Layer Perceptron (MLP)?
A Multi-Layer Perceptron is a neural network with multiple hidden layers

between input and output layers.

Used for:

Classification

Regression

Pattern recognition

16. Features of Deep Learning

Automatic feature extraction

Handles large datasets

High accuracy

Works well with unstructured data (images, text, audio)

Requires powerful hardware (GPU)

17. When to Use Which Activation Function?

Activation Function	When to Use
ReLU	Hidden layers in deep networks
Sigmoid	Binary classification output
Softmax	Multi-class classification
Tanh	Hidden layers when negative values are needed

18. What is FNN? When to Use It?

FNN (Feedforward Neural Network) is a neural network where data flows only in one direction (input ⟶ output).

Use when:
Data is structured
No sequential dependency

Example:
Tabular data prediction

19. What is CNN?
CNN (Convolutional Neural Network) is a deep learning model designed for image processing.

Used for:
Image classification
Object detection
Facial recognition

20. What is LSTM?
LSTM (Long Short-Term Memory) is a special type of RNN designed to remember long-term dependencies in sequence data.

Used in:
Speech recognition
Language translation

Time series forecasting

21. What is RNN?

RNN (Recurrent Neural Network) is used for sequential data where previous information influences future outputs.

Example:

Text generation

Speech recognition

22. What is GAN?

GAN (Generative Adversarial Network) consists of two neural networks:

Generator

Discriminator

They compete to generate realistic data.

Used for:

Image generation

Deepfakes

Data augmentation

23. What is GRU?

GRU (Gated Recurrent Unit) is a simplified version of LSTM used for sequential data.

Advantages:
Faster training
Fewer parameters

24. What is Transformer?
A Transformer is a deep learning model that uses self-attention mechanisms instead of recurrence.

Used in:
Language models
Machine translation
Chatbots

Example models:
BERT
GPT